DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Winter Examination – 2023

Course: B. Tech. Branch: E&TC Engineering Semester: VI

Subject Code & Name: BTETPE604A_Y18 & CMOS Design

Max Marks: 60 Date: 23/1/2024 Duration: 2:00 to 5:00 PM

Instructions to the Students:

1. *All the questions are compulsory.*
2. *The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.*

		(CO)	Marks
Q. 1	**Solve Any Two of the following.**		**12**
A)	**Implement D flip-flop using static CMOS logic**	**1**	**6**
B)	**Implement $Y = (A + B).(\bar{C} + \bar{D})$ function using 2 input NOR gates only & then draw the schematic using static CMOS design style. Assume inputs available only in true form.**	**2**	**6**
C)	**Draw the schematic and layout for three input CMOS NOR gate, Use CMOS Lambda based design rules.**	**2**	**6**
Q.2	**Solve Any Two of the following.**		**12**
A)	**Write a short note on CMOS scaling**	**1**	**6**
B)	**Explain leakage power in CMOS inverter. What are ways to reduce it?**	**1**	**6**
C)	**Implement the full adder using half adder.**	**2**	**6**
Q. 3	**Solve Any Two of the following.**		**12**
A)	**What are various Short Channel Effects? How to avoid them?**	**1**	**6**
B)	**Write electrical, logical effort, parasitic effort for four input NOR gate.**	**2**	**6**
C)	**Explain CMOS λ-based design rules in detail.**	**1**	**6**
Q.4	**Solve Any Two of the following.**		**12**
A)	**Implement T Flip-Flop using Static CMOS design style.**	**2**	**6**
B)	**Explain CMOS Domino logic in detail. Compare it with static CMOS logic**	**2**	**6**
C)	**Explain all the parasitic capacitances of the MOSFET in detail**	**1**	**6**
Q. 5	**Solve Any Two of the following.**		**12**
A)	**Implement 2-bit magnitude comparator using static CMOS logic**	**1**	**6**
B)	**Implement half adder combinational circuit using static CMOS logic.**	**2**	**6**
C)	**Implement $\overline{AB + CD}$ gate using dynamic CMOS logic.**	**2**	**6**

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